Application Report
September 2020

Jacinto7 DDRSS Tools – User Guide

Jacinto Processors *Texas Instruments*

ABSTRACT

DOCUMENT HISTORY

Version	Date	Author	Notes
Beta	September 17, 2020	Texas Instruments Incorporated	

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1 OVERVIEW

This document describes how to use the Beta version of the Jacinto7 DDRSS margin analysis tool.

1.1 Components of Package

- Pre-built executable binary in the "bin" folder
- CCS project files in the "build" folder
- Documentation and example in the "docs" folder. Examples in the "example" folder include:
    - Output of binary executable run on the DUT
    - Output of post-processing script
- GEL file in "gel" folder to
    - Configure the PINMUX as UART IO
    - Configure the R5 RAT
- Library binaries and header files used by source code in the "lib" folder
- Post-processing application in the "perl" folder
- Source code (excluding libraries) of the DUT executable in the "src" folder

1.2 Supported Features / Limitations

The following lists are not limited to, but hi-light supported features and limitations.

<u>Supported Features</u>

- 2-dimensional plots of the WRITE and READ DQ bits
    - Vertical access represents reference voltage
    - Horizontal access represents DLL delay value
- Plots provided per DQ bit
- Plots provided for both edges of DQS
- Plots color coded to illustrate:
    - Trained value
    - Passing region
    - Gradient color representing failure rate
- Auto detection of bus-width for optimized pattern generation
- Custom run-time step size (for both voltage and delay) to allow flexibility between reduced test time versus greater precision
- UART output to decrease test time

<u>Limitations</u>

- The DUT software can only execute either the read or write code before requiring a re-configuration of the DDRSS.
- Axis values illustrated in raw values instead of voltage / approximate time
- VREF minimum step size limited to 2, delay minimum step size limited to 8
- Test pattern fixed
- Traffic initiator (R5) fixed
- Requires Code Composer Studio / JTAG

- Post-processing script tested on Windows operating system only
- Provided GEL file only supports PINMUX functions for UART0

2 LPDDR4 MARGIN ANALYSIS TOOL

The following section describes how to use the Jacinto7 DDRSS margin analysis tool.

2.1 Preliminary Requirements

The following requirements are necessary to use the LPDDR4 margin analysis tool:

- Code Composer Studio (version 10.1.0 or later)
- Emulator / JTAG access
- On board UART access for data logging
- DDR configuration (executed prior to using tool)
- Perl
- Microsoft Excel

2.2 Executing the Software on the DUT

The following section provides high-level instructions on executing the software on the device under test.

1. Import the CCS project and re-build with the desired UART interface accessible by a PC
    a. Modify the value of *UART_INSTANCE* in the source file *jacinto7_ddrss_tools_main.c* to configure the desired UART interface
2. Boot the DUT and configure the DDR interface (ex: use a GEL file)
3. **IMPORTANT**: Connect to the **MCU R5** core and load the accompanying GEL file
4. Run the appropriate scripts from the GEL to
    a. Configure the appropriate PINMUX to UART. User must add necessary code if not using UART0.
    b. Configure the R5 RAT.
5. Load the executable from the "bin" folder to the **MCU R5** core.
6. Open up a terminal window (ex: teraterm) and connect to the appropriate COM port with the following settings: 115200 baud rate, 8bit data, no parity, 1bit stop.
7. Log the terminal output using the terminal software logging feature (ex: *File→Log* for teraterm). Any unique file name is valid.
8. Run the executable and follow the prompts from the terminal window
    a. Select either read or write
    b. Specify desired VREF step size (ex: 4)
    c. Specify desired delay step size (ex: 16)
9. Upon completion of test code, close the terminal log

2.3 Using the Post-Processing Script

The following section provides high-level instructions on executing the post-processing script.

1. From the command line, execute the Perl script found in the "perl" folder, passing the log file saved from section "Executing the Software on the DUT" as an input parameter.

a. Example: *jacinto7_ddrss_tools_margin_Analysis.pl ../docs/example/LP4_Read_DlyStep16_VrefStep4_Example.txt*
b. The post-processing time will vary depending on the delay step sizes. The script updates progress while generating the results.

2.4 Viewing and Interpreting Results

Results of the post-processing script will be provided in a Microsoft Excel workbook, found in the same directory as the input file.

Figure 1 illustrates an example of the expected output. A plot is provided for both the rising and falling edge of DQS.

Cells colored purple represent the approximate location of the trained value for the corresponding DQ bit. In some cases, more than one cell will be colored purple if the trained value does not match the available axis values. In this case, the two closest axis values (one greater and one less than) compared to the trained value will both be colored purple. A maximum of four cells could be colored purple.

Cells colored green represent the passing region for the corresponding DQ bit.

All other cells represent the failing region, using a gradient color scheme to illustrate failure rate. Cells colored red represent the highest failure rate, while cells colored blue represent the least failure rate.

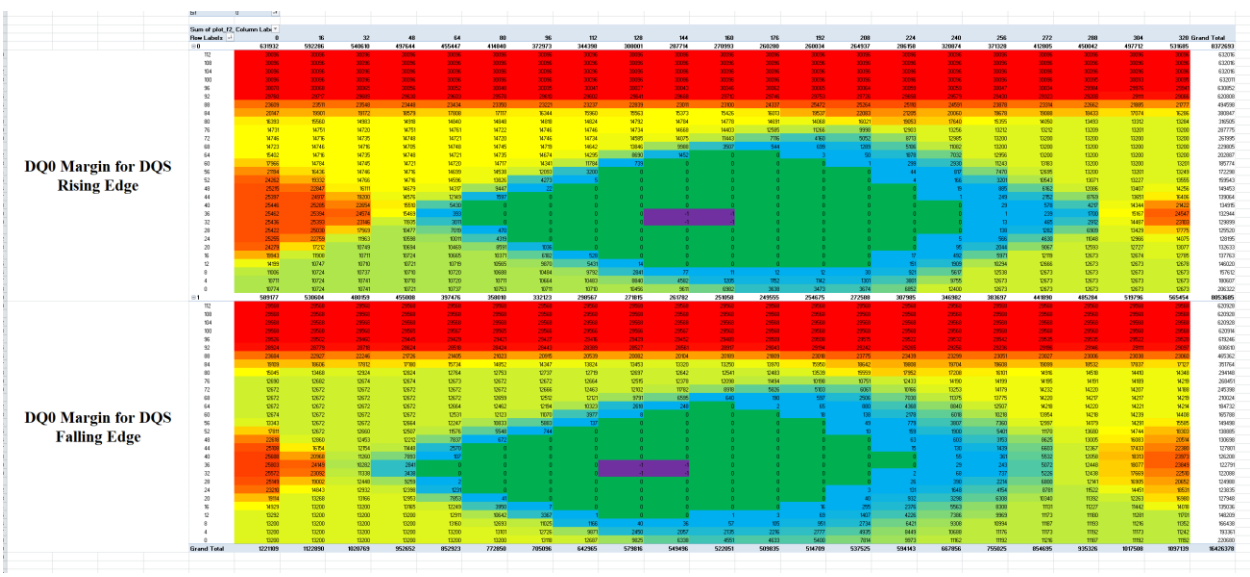

Figure 1 - Example READ Plot

Each DQ plot is provided on a separate tab, as illustrated below in Figure 2.

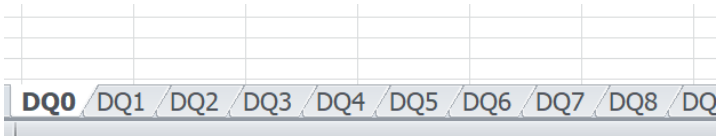

Figure 2 - Results per DQ Bit

3 REFERENCES

[1]

4 REVISION HISTORY

The revision history highlights any major changes made to this document since the previous revision.

Location	Additions / Modifications / Deletions